

IT2120

Introduction to Statistics

Lecture 01 - Easy Exam Notes

Covers the full 28-slide lecture

HOW TO USE THIS NOTE Read the short explanation first, then learn the EXAM IMPORTANT boxes and MEMORY TIP boxes. The final pages give a fast revision checklist.

MAIN EXAM TARGETS Population vs sample, parameter vs statistic, variable types, measurement scales, descriptive vs inferential statistics, and the lecture example.

SLIIT - Year 02 Semester 01

Source: 2026-S2-IT2120-Lecture-01-IntroductionToStatistics.pdf

Lecture Map

Lecture coverage: slides 1-28

What the lecture covers

Slides 1-5	Introduction, applications, definitions, uncertainty and variability
Slides 6-14	Core terminology: population, sample, variable, data, parameter, statistic, experiment + worked example
Slides 15-19	Types of variables: qualitative/quantitative, nominal/ordinal, discrete/continuous
Slides 20-21	Measurement scales and interval vs ratio comparison
Slides 22-24	Areas of statistics: descriptive and inferential
Slides 25-27	Statistical packages and introduction to R
Slide 28	End of lecture / questions

MEMORY TIP Think of the lecture as 5 blocks: Meaning -> Terminology -> Variable Types -> Measurement Scales -> Areas + R.

1. What is Statistics?

Lecture coverage: slides 3-5

Applications of Statistics (Slide 3)

Statistics can be applied in many fields. The lecture examples are:

- Engineering and Sciences
- Medical Sciences
- Education
- Business Analytics
- Social Sciences
- Machine Learning
- Quality Control
- Actuarial Sciences

Definition and Main Idea (Slides 4-5)

- Statistics is the study of the collection, analysis, interpretation, presentation, and organization of data.
- Statistics can also be viewed as numbers that summarize raw facts and figures in a meaningful way.
- The lecture also describes statistics as the study of uncertainty.
- We need statistics to identify variability in data - values are not always identical, so statistics helps us understand the differences.

EXAM IMPORTANT If asked "What is statistics?", mention the data process: collect -> analyze -> interpret -> present -> organize. Also remember uncertainty and variability.

MEMORY TIP CAIPO = Collection, Analysis, Interpretation, Presentation, Organization.

2. Core Statistical Terminology

Lecture coverage: slides 6-13

Population, Census, Sample and Sample Survey

Population	The complete collection of individuals or objects that the researcher is interested in. A population may be finite or infinite.
Census	Data are collected from all individuals in the population.
Sample	A subset (part) of the population.
Sample survey	Data are collected only from a sample, not from the whole population.

MEMORY TIP Census = complete population. Sample survey = some of the population.

Variable and Data

Variable	A characteristic/property of each individual in a population or sample. Examples in the lecture: Age, Gender, Temperature. Variables are usually denoted by uppercase letters.
Data (singular - lecture wording)	The value of a variable for one element of a population or sample. The value may be a number, a word, or a symbol.

EXAMPLE If the variable is Age, one student may have the data value 21. If the variable is Gender, one data value may be a word/category.

Parameter vs Statistic - Very Important

Point	Parameter	Statistic
Related to	Population	Sample
Meaning	Summary characteristic of population individuals	Summary characteristic of sample individuals
Mean	Population mean (μ , μ)	Sample mean ($\bar{x}$)
Variance	Population variance (σ^2 , σ^2)	Sample variance (S^2)
Proportion	Population proportion (P)	Sample proportion (p)

EXAM IMPORTANT Population $\rightarrow$ Parameter. Sample $\rightarrow$ Statistic. This relationship is one of the most important definitions in the lecture.

MEMORY TIP P-P and S-S: Population-Parameter; Sample-Statistic.

Experiment

- An experiment is a planned activity whose results yield a set of data.

3. Worked Lecture Example

Lecture coverage: slides 14

Average weight of a first-year student at SLIIT

The researcher wants the average weight of a first-year student in SLIIT, but collected data from all first-year students in the Computing Faculty.

Population	All first-year students in SLIIT
Sample	All first-year students in the Computing Faculty
Variable	Weight
Summary characteristic	Average weight -> Statistic
Type of survey	Sample survey

WHY IS IT A STATISTIC? The average is calculated from the sample (Computing Faculty students), not from all first-year students in SLIIT. Therefore the summary is a statistic.

WHY IS IT A SAMPLE SURVEY? Only part of the population is observed. If data were collected from every first-year student in SLIIT, it would be a census.

EXAM METHOD For similar questions, identify in this order: 1) target group = population, 2) observed group = sample, 3) measured property = variable, 4) summary from population/sample = parameter/statistic, 5) all or part = census/sample survey.

4. Types of Variables

Lecture coverage: slides 15-19

Main Classification (Slide 16)

TYPES OF VARIABLES			
Qualitative Variables		Quantitative Variables	
Nominal	Ordinal	Continuous	Discrete

Qualitative vs Quantitative

Qualitative / Attribute / Categorical	Categorizes and describes an element. Lecture examples: Hair color, Gender, Marital status, Highest education qualification.
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Quantitative / Numerical

Quantifies an element. Lecture examples: Marks for statistics, Age, Temperature, Time taken to travel to SLIIT from home.

MEMORY TIP Qualitative = quality/category. Quantitative = quantity/number.

Qualitative Variables: Nominal vs Ordinal

Type	Meaning	Lecture examples
Nominal	Categories are not naturally ordered.	Gender, Hair Color, Marital Status
Ordinal	Categories are naturally ordered.	Satisfaction Rating, Pain Severity, Highest education qualification

FAST TEST Can the categories be ranked naturally? Yes -> Ordinal. No -> Nominal.

Quantitative Variables: Discrete vs Continuous

Type	Lecture meaning	Lecture examples
Discrete	Distance/gap between two values exists. Think of separate countable values.	Age in years; number of children in a family; number of accidents at a junction within an hour
Continuous	Can contain any value within a given range.	Temperature; heart beat of a patient

MEMORY TIP Qualitative -> NO (Nominal, Ordinal). Quantitative -> DC (Discrete, Continuous).

5. Measurement Scales

Lecture coverage: slides 20-21

Four Levels: Nominal -> Ordinal -> Interval -> Ratio

The lecture diagram places the scales from the lowest/weakest measurement to the highest/strongest measurement:

Nominal Data	Categories with no ordering or direction. Lowest level / weakest form of measurement.
Ordinal Data	Ordered categories - ranking, order or scaling.
Interval Data	Differences between measurements can be evaluated, but there is no true zero.
Ratio Data	Differences between measurements can be evaluated and a true zero exists. Highest level / strongest form of measurement.

MEMORY TIP NOIR = Nominal, Ordinal, Interval, Ratio. Easy ladder: Name -> Order -> Interval -> Real zero.

Interval Scale vs Ratio Scale - Lecture Comparison

Feature	Interval Scale	Ratio Scale
Operations	Variables can be added and subtracted; ratio and multiplication are not possible.	Includes ratio and multiplication characteristics in addition to interval-scale characteristics.
Center measures	Mean, median and mode can be calculated.	Mean, median and mode can be calculated.
Differences	Differences between variables can be evaluated.	Differences between variables can be evaluated.
True zero	No true zero point.	True zero point exists.
Signs	Can consider positive and negative numbers.	Lecture states only positive numbers.
Examples	Temperature in Celsius, Temperature in Fahrenheit, pH Value	Height, Weight, Temperature in Kelvin, Number of sales, Income of an individual, Heart Rate

EXAM IMPORTANT The easiest distinction is the true zero: Interval = no true zero; Ratio = true zero. Learn the lecture examples because identification questions may use them.

LECTURE-SPECIFIC NOTE The table above follows the wording and examples shown in the provided lecture slide. For exam revision, use the lecturer's definitions and examples.

6. Areas of Statistics

Lecture coverage: slides 22-24

Statistics -> Descriptive + Inferential

Descriptive Statistics	Also called preliminary analysis / explanatory analysis in the lecture. Gives a rough idea about the behavior of data and describes how variables behave.
Methods under Descriptive	1) Graphical Methods 2) Numerical Methods
Inferential Statistics	Draws conclusions about population parameters by using sample statistics.
Main areas under Inferential	1) Parameter Estimation 2) Hypothesis Testing

MEMORY TIP Descriptive DESCRIBES the data you have. Inferential INFERS about the population from a sample.

EXAM IMPORTANT Be able to state both branches and the two sub-areas of inferential statistics: parameter estimation and hypothesis testing.

7. Statistical Packages and R

Lecture coverage: slides 25-27

Statistical Packages

- Statistical packages help analyze data easily and precisely.
- Common packages listed in the lecture: SPSS, SAS, Minitab, R, E-views and Matlab.
- This module uses R for data analysis.

Introduction to R Software

Nature	Independent and open source.
Origin	Initially developed at the University of Auckland in the mid-1990s.
License	Distributed under the GNU open software license.
Development	Developed by the user community.
Platforms	Linux, Windows and Mac.
Version in lecture	4.4.2 - released in 2024.
Interfaces	Terminal and GUI available.
IDEs listed	R Studio, Rattle.

EXAM TIP For a short-answer question on R, remember: open source, Auckland, GNU, community-developed, multi-platform, terminal/GUI, R Studio/Rattle.

8. Exam-Focused Quick Revision

Lecture coverage: slides Full lecture

Must-Know Differences

Pair	First	Second
Population vs Sample	Whole target group	Subset of the population
Census vs Sample Survey	Collect from all population members	Collect from part/sample
Parameter vs Statistic	Population summary	Sample summary
Qualitative vs Quantitative	Category/description	Numerical quantity
Nominal vs Ordinal	No natural order	Natural order
Discrete vs Continuous	Separate values / gaps	Any value within a range
Interval vs Ratio	No true zero	True zero
Descriptive vs Inferential	Describe observed data	Conclude about population using sample

Likely Exam Tasks

- Define statistics, population, sample, variable, data, parameter, statistic, and experiment.
- Given a scenario, identify population, sample, variable, parameter/statistic, and survey type.
- Classify a variable as qualitative or quantitative, then into nominal/ordinal/discrete/continuous.
- Identify a measurement scale: nominal, ordinal, interval, or ratio.
- Compare interval and ratio scales, especially true zero and examples.
- Explain descriptive vs inferential statistics and name their methods/sub-areas.
- List common statistical packages and key features of R from the lecture.

One-Minute Memory Sheet

Statistics process	Collect -> Analyze -> Interpret -> Present -> Organize
Population/Sample	Population = whole; Sample = part
Summary	Population -> Parameter; Sample -> Statistic
Survey	All -> Census; Part -> Sample survey
Variable types	Qualitative -> Nominal/Ordinal; Quantitative -> Discrete/Continuous
Scales	NOIR: Nominal -> Ordinal -> Interval -> Ratio
True zero	Interval = no; Ratio = yes
Areas	Descriptive = describe; Inferential = infer
Inferential sub-areas	Parameter estimation + Hypothesis testing
R	Open source; Auckland; GNU; Linux/Windows/Mac; R Studio/Rattle

FINAL EXAM TIP When a question gives a story, do not rush. First ask: Who is the full target group? Who was actually observed? What property was measured? Then classify the summary and survey type.

9. Full Slide Coverage Checklist

Lecture coverage: slides 1-28

- Slides 1-2: Module and lecture title
- Slide 3: Applications of statistics
- Slides 4-5: Definitions, uncertainty, variability
- Slides 6-13: Terminologies and definitions
- Slide 14: Worked population/sample/statistic example
- Slides 15-19: Types of variables and examples
- Slides 20-21: Measurement scales; interval vs ratio
- Slides 22-24: Areas of statistics
- Slide 25: Statistical packages and use of R
- Slides 26-27: Introduction and features of R
- Slide 28: End of lecture / questions

READY TO REVISE If you can explain every row in the one-minute memory sheet without looking, you have covered the core exam content of this lecture.